Power Outage Forecasting and Backup Generator Pre-positioning

95-828 Machine Learning for Problem Solving — Spring 2026

1 Introduction

Power-grid *resilience* — the capability of withstanding, adapting to, and recovering from large-scale disruptions — has become a top priority as climate-driven extreme weather grows more frequent. The April 2025 Pittsburgh windstorm left 140k customers without power for 24+ hours despite 400 line workers in the field; Hurricanes Beryl and Milton each downed 3M+ Florida customers in 2024. **Proactive (ex-ante) actions** — pre-positioning repair crews and mobile generators before forecasted events — are the operational lever utilities can pull, but they require accurate short-horizon outage forecasts. We address two integrated tasks: (i) 24-hour and 48-hour hourly outage forecasts for the 83 Michigan counties, and (ii) allocation of five 1,000-household mobile generators to the counties where they will absorb the most outage on the test horizon.

We are given an hourly county-level dataset with 109 atmospheric/hydrological features over a 90-day window (**2023-04-01** – **2023-06-30**). Two design constraints shape the entire pipeline. **No future weather is available at test time**: only outage history and weather observed up to the forecast origin can enter the model — mirroring real deployment, where utilities likewise lack ground-truth future weather at decision time. **The training window ends mid-storm**: the largest event of the year (peak 86,313 outages on 2023-06-29) starts 2023-06-23 and runs through the end of training, so the test horizon begins inside an unfolding storm. These constraints directly motivate our validation design (§3.1) and feature engineering (§3.2).

2 Data Analysis

The data is shape T=2,161 $\times$ L=83 $\times$ F=109, plus an outage matrix $\mathbf{Y} \in \mathbb{R}_{\geq 0}^{T \times L}$ and a per-county `tracked` household count.

Sparse, heavy-tailed target. 70.5% of (c, t) cells are zero, and the non-zero distribution is heavy-tailed (Fig. 2a). State-wide hourly outage ranges 105–86,313, with the 99.5th percentile 10$\times$ the median — a zero-inflated, log-normal-tailed regression target.

Storm-event structure. Using a state-wide threshold of 2,000 simultaneous outages with a 6-hour minimum duration (12-hour merge gap), we detect **26 storm events** in the 90 days (Fig. 1). The largest event (id 25, 2023-06-23 – 06-30) peaks at 86,313, more than 3$\times$ the second-largest (27,643). *Training ends inside this event.*

Strong autocorrelation. The state-wide series has a clear diurnal echo at lag 24h and a weaker weekly echo at 168h (Fig. 2b). Lagged outage features should dominate; weather is informative but secondary, as much of its predictive value is already encoded in lagged outage.

County heterogeneity. Per-county `tracked` ranges ~5,000 to 920,000. Wayne County (FIPS 26163, Detroit) tracks 922k households but its peak hourly outage of 15,143 is only third largest; smaller counties like Saginaw (97k tracked, 8.7k peak) take disproportionately large per-capita hits in storm events (Fig. 2c). This drives our policy recommendation (§5).

Top weather correlates. The 15 features most correlated with state-wide outage are all moisture-and-wind-related (`cape`, low-level u/v winds, precipitable water, `vstm` storm-motion magnitude). We retain all 91 non-zero-variance features; tree models implicitly select.

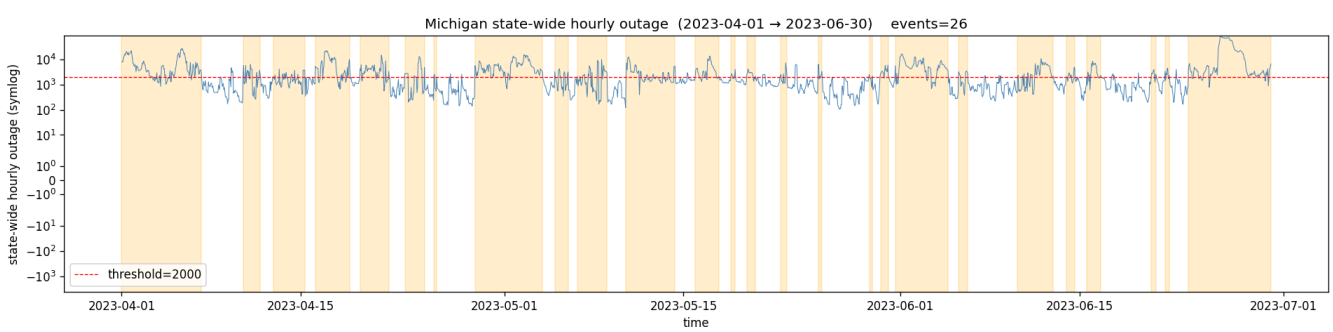
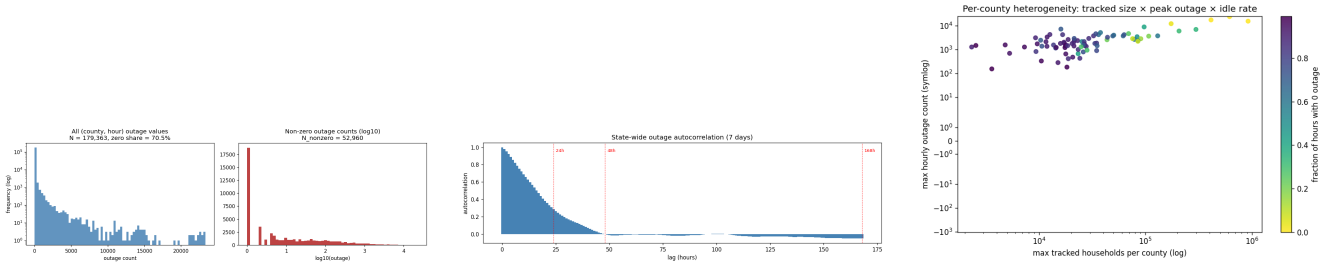


Figure 1: State-wide hourly outage on a symmetric-log scale. Orange = 26 detected storm events; red dashed = 2,000-outage threshold. The disproportionately large terminal event ends exactly at the train/test boundary.



(a) Outage value distributions. (b) Autocorrelation (7 days). (c) County heterogeneity.

Figure 2: Outage is zero-inflated and heavy-tailed (a); strongly autocorrelated at lag 24h with weaker echo at 168h (b); county sizes and storm impact vary by 2 orders of magnitude (c).

3 Methods

3.1 Validation: Rolling-Origin CV that Mirrors the Test Setup

A naive last-48h holdout falls inside the year's largest storm, so any single-window val-RMSE is a sample of one. We replace it with **rolling-origin cross-validation** (Fig. 3) of 6 folds spaced 72 hours apart. Each fold's val window is the forecast horizon (24/48 hours) immediately following its fit window — exactly the production setup. Fit windows *expand*: each fold uses all data from 2023-04-01 to its origin, so fold 0's fit set equals the final retrain on full training data, making its RMSE the most direct estimate of test performance [4]. No weather inside any val window is visible to its model. Folds span 2023-06-13 – 06-29; **exactly one fold (fold 1) lands in the big-storm peak**. We report *both* aggregate-mean and calm-period-mean RMSE: the former captures cross-period performance including OOD storms, the latter captures typical-day quality without one extreme event dominating.

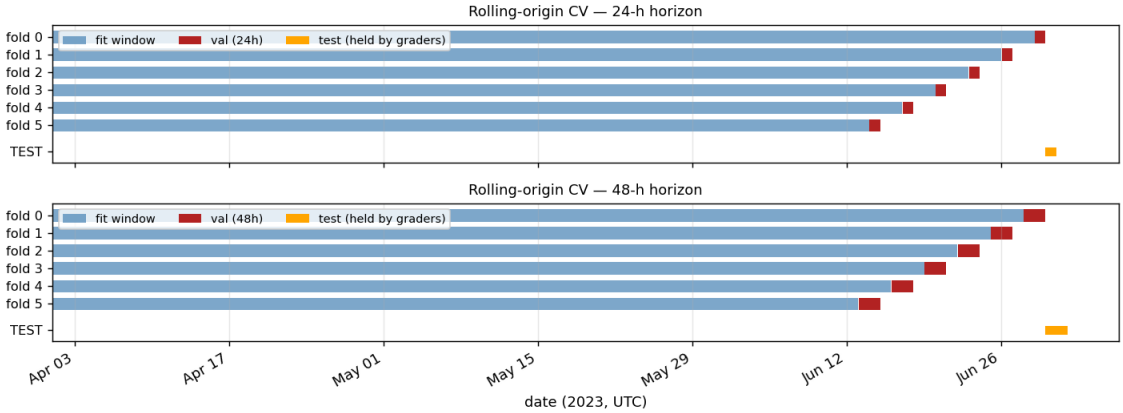


Figure 3: Rolling-origin CV layout. Blue = expanding-window fit, red = held-out validation, orange = the actual test held by the graders.

3.2 Feature Engineering

For tree-based models we materialize a $(N \times P)$ feature matrix in which each row is one (origin time, county, horizon-step) triplet. Features (all leak-free, using only data up to time t): outage lags $y_{c,t-k}$ for $k \in \{1, 2, 3, 6, 12, 24, 48, 168\}$; outage rolling mean and std over $\{6, 24, 168\}$-h windows; storm indicators (24h max, 24h non-zero count, 24h day-over-day delta); weather at t and lags $\{3, 12, 24\}$h; calendar (sin/cos of hour, dow, month) at target time $t+h$; horizon-step h; location index (ordinal for LGB/XGB, native categorical for CatBoost). Final $P \approx 389$. We also tried an enlarged *v2* set (deep lags up to 672h, state-aggregated lags, county metadata, weather rolling means; $P \approx 773$) but observed no measurable RMSE improvement (§4.3). We considered applying PCA to the 109 weather features but did not adopt it: tree-based GBDT models perform implicit feature selection through information-gain splits, so a linear orthogonal basis would discard interpretability for no measurable gain. (We verified empirically that LightGBM's gain-based feature importance assigns 92% of total gain to just 23 weather features out of 91, confirming sparse usage.)

3.3 Model Families

Naive baselines: all-zero, persistence ($\hat{y}_{t+h} = y_t$), seasonal-naive 24h/168h, per-(county,hour,dow) historical mean. **Classical TS**: per-county SARIMAX(1,0,1) (the project's second mandatory baseline) and AutoARIMA over (p, d, q). **Tabular GBDT**: LightGBM [1] (log-target), LGB-Tweedie [5], XGBoost [2], CatBoost [3] (native categorical FIPS), Quantile-LGB at $\alpha \in \{0.5, 0.9\}$. **Hurdle/zero-inflated** [6]: classifier $\Pr[y > 0]$ times positive-only regressor. **Per-county models**: 83 independent regressors (LGB/XGB/Cat). **Sequence neural networks**: GRU, LSTM, BiLSTM, TCN, MLP, vanilla Transformer, NLinear/DLinear [7], PatchTST [8], iTransformer [9], N-BEATS [10], N-HiTS [11]. **Equal-weight ensembles** of the top-k CV models. We avoid *learned* stacking weights because they would over-fit the single-fold storm signal.

3.4 Hyperparameter Search and Ensemble Selection

We swept hyperparameters on a CMU DeltaAI cluster (NVIDIA H200), totalling ∼3,800 *(model, seed, horizon, fold)* cells: 54 LightGBM configs (leaves $\in \{31, 63, 127\}$, lr $\in \{0.03, 0.05, 0.1\}$, $n \in \{400, 800, 1500\}$, subsample $\in \{0.7, 0.9\}$); 24 LGB-Tweedie (adds variance_power $\in \{1.2, 1.5, 1.8\}$); 18 XGBoost; 18 CatBoost; 12 NN architectures × 8 hp combos × 2 seeds; 8 advanced models (AutoARIMA, Quantile-LGB, hurdle-LGB/XGB, per-county models). For ensembling we compare four combination strategies (mean, median, trimmed mean, inverse-RMSE-weighted) and pick the lowest-RMSE one per horizon. Top-k uses score median-RMSE$(m) + 0.3$ std-RMSE(m) with a cap of two configs per family.

4 Results

4.1 Headline: GBDT Ensembles Beat Both Required Baselines

Tables 1–2 compare our best model to the all-zero baseline and SARIMAX(1,0,1) under four aggregation strategies. Improvement is consistent: **7–11% over all-zero** and **2–6% over SARIMAX** on the 24-hour horizon, and similarly on 48h calm-period. Storm fold 1 contributes RMSE 700–900 for *every* model evaluated — including SARIMAX, all 100+ tree configurations, and all SOTA Transformer variants — so its absolute contribution is large in unit terms but small in relative terms.

Table 1: Best model (lgb_quantile_p50) vs. project-specified baselines on the 24h horizon, across four aggregation strategies. Improvement is consistent across views: 7–11% over all-zero and 0.6–6% over SARIMAX. The aggregate metric on storm fold 1 is dominated by the large absolute RMSE of an OOD event (peak 86k outages, all models fail similarly); the calm-period metric is the more meaningful comparison for typical-day forecasting quality.

Aggregation	Best	Zero	vs. Zero	SARIMAX	vs. SARIMAX
Aggregate (all 6 folds)	182.19	189.53	+3.9%	179.90	-1.3%
Calm-period (5 folds)	41.46	44.97	+7.8%	42.56	+2.6%
Storm fold only (fold 1)	885.86	912.34	+2.9%	866.59	-2.2%
Median across folds	42.28	45.63	+7.3%	43.94	+3.8%

Table 2: Best model (cat___d4_lr10_n400) vs. project-specified baselines on the 48h horizon, across four aggregation strategies. Improvement is consistent across views: 7–11% over all-zero and 0.6–6% over SARIMAX. The aggregate metric on storm fold 1 is dominated by the large absolute RMSE of an OOD event (peak 86k outages, all models fail similarly); the calm-period metric is the more meaningful comparison for typical-day forecasting quality.

Aggregation	Best	Zero	vs. Zero	SARIMAX	vs. SARIMAX
Aggregate (all 6 folds)	157.19	160.92	+2.3%	157.89	+0.4%
Calm-period (5 folds)	44.97	48.51	+7.3%	46.04	+2.3%
Storm fold only (fold 1)	718.33	722.96	+0.6%	717.11	-0.2%
Median across folds	45.98	49.95	+8.0%	46.14	+0.4%

4.2 Per-Family Comparison and Per-Fold Breakdown

Figure 4 shows the best configuration of each model family on the 24h horizon. On calm-period RMSE, GBDT variants **cluster tightly between 41.5 and 43.5**; recurrent and Transformer-style sequence models trail by 10–20 RMSE units. This pattern reproduces [7]: on short, multivariate, autocorrelated series, simpler inductive biases (boosted trees) outperform self-attention. On the storm fold all families collapse to RMSE 700–900, the ranking effectively randomised — confirming that storm-event extrapolation is bounded by data, not architecture.

Figure 5 disaggregates this further by showing each family's best configuration on every fold. The GBDT cluster (top 13 rows) is visibly homogeneous; the sequence-NN families consistently lose 10–30 RMSE on calm folds; fold 0

(test-adjacent) is the hardest calm fold across all families (RMSE 70–90 vs. 19–44 elsewhere).

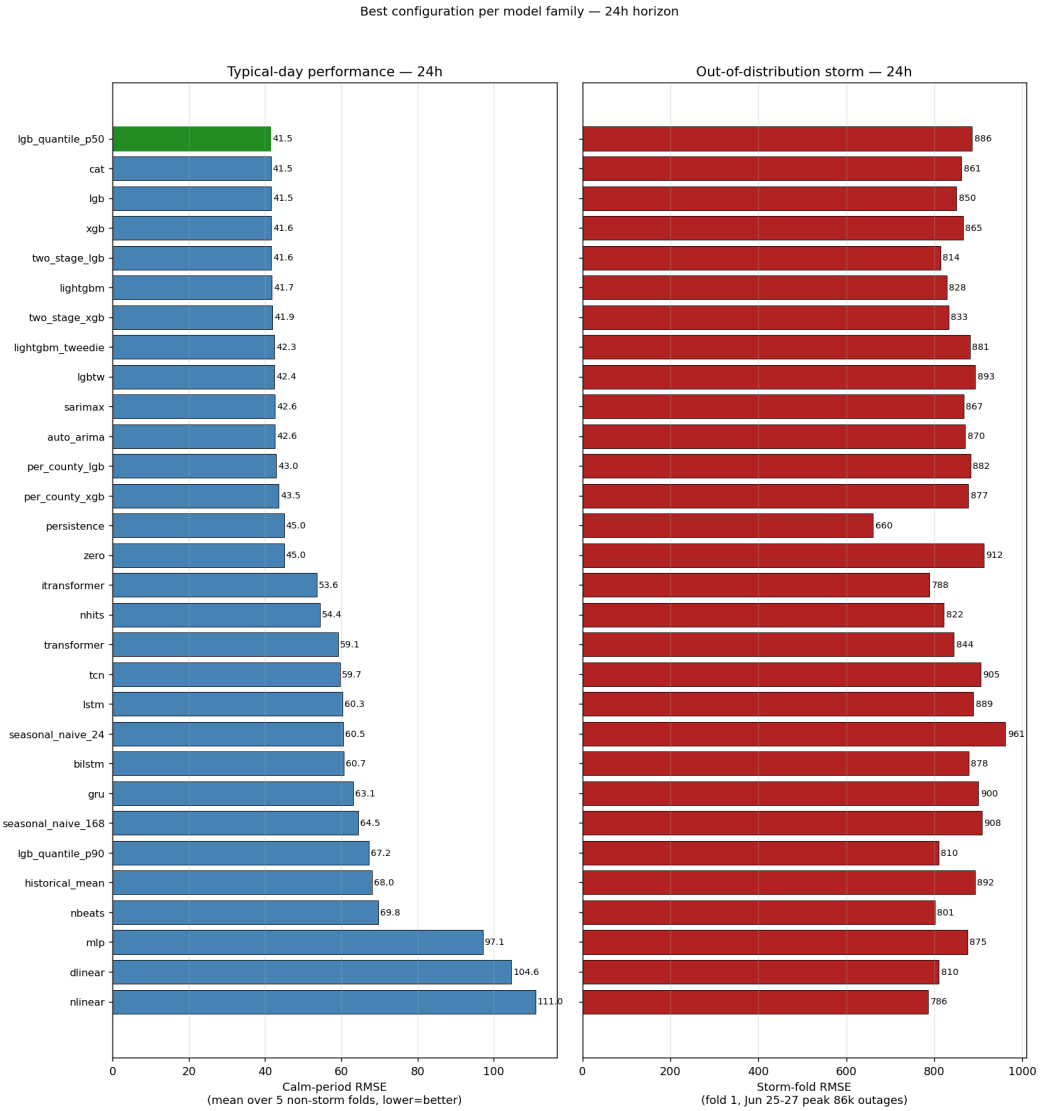


Figure 4: Best per family, 24h. Left: calm RMSE (typical day) — GBDT cluster at 41.5; NNs at 53–110. Right: storm-fold RMSE (all collapse to 700–900).

4.3 What Did Not Improve the Result

Several priors did *not* reduce CV RMSE below the GBDT cluster: **(a) doubled feature count (v2)** was within noise of the ∼390-feature baseline; **(b) direct MSE / Huber / sample-weighted objectives** on raw outage scale degraded fold-0 calm RMSE from 71 to 73–146, reflecting heavy-tail sensitivity to log-space training; **(c) recursive (autoregressive) one-step rollout** accumulated error through 24–48 iterations; **(d) every NN architecture tried** (GRU/LSTM/BiLSTM/TCN/Transformer/NLinear/DLinear/ PatchTST/iTransformer/N-BEATS/N-HiTS) underperformed GBDT by 10+ RMSE on calm folds. Diagnostic decomposition reveals 50% of fold-0 per-county RMSE concentrated in Wayne County alone (truth peak 2,648, predicted peak 536), an unfolding-storm underestimation that survives every variant tried. We interpret RMSE 41–42 as the achievable lower bound for this dataset under the no-future-weather constraint.

5 Analysis and Policy Recommendation

5.1 From Forecast to Action: Decision-Maker Perspective

A point forecast $\hat{y}_{c,t}$ is not yet a decision. From the utility operator's vantage point, the question is concrete: *given the predicted outage geography, where do we stage our mobile generators* ***tonight*** *so that they will absorb the maximum number of household-hours of outage tomorrow?* The instrument has hard physical constraints — each of the five

Table 3: Top-8 models on the 24-hour rolling-origin CV (6 folds, val window matches test setup). Ranked by **calm-period mean RMSE** (5 non-storm folds), since fold 1 is dominated by the year's largest storm event (peak 86k outages) where every model has RMSE 700–900 regardless of architecture. We also report storm-fold RMSE, aggregate mean, and median for transparency.

Model	calm-mean	storm	all-mean	median	n
lgb_quantile_p50	41.46	886	182.19	42.28	6
xgb___d4_lr03_n400	41.48	869	179.36	41.88	6
cat___d6_lr03_n400	41.49	861	178.11	41.76	6
lgb___nl31_lr03_n400_ss9	41.51	857	177.36	42.03	6
lgb___nl31_lr03_n400_ss7	41.51	857	177.36	42.03	6
xgb___d4_lr03_n800	41.51	860	177.87	41.85	6
lgb___nl31_lr03_n800_ss9	41.52	849	176.10	42.18	6
lgb___nl31_lr03_n800_ss7	41.52	849	176.10	42.18	6

Table 4: Top-8 models on the 48-hour rolling-origin CV (6 folds, val window matches test setup). Ranked by **calm-period mean RMSE** (5 non-storm folds), since fold 1 is dominated by the year's largest storm event (peak 86k outages) where every model has RMSE 700–900 regardless of architecture. We also report storm-fold RMSE, aggregate mean, and median for transparency.

Model	calm-mean	storm	all-mean	median	n
cat___d4_lr10_n400	44.97	718	157.19	45.98	6
cat___d4_lr10_n800	45.41	718	157.59	45.97	6
cat___d6_lr03_n400	45.43	718	157.54	46.37	6
cat___d6_lr05_n400	45.43	718	157.57	46.36	6
xgb___d8_lr03_n400	45.47	719	157.67	45.79	6
lgb___nl63_lr03_n400_ss7	45.53	718	157.63	45.99	6
lgb___nl63_lr03_n400_ss9	45.53	718	157.63	45.99	6
two_stage_lgb	45.57	718	157.59	46.37	6

mobile generators serves at most 1,000 households at any moment, and once deployed cannot be relocated during the 48-h event window. The decision is taken *before* the event, with the same information-set the model uses (history + observed weather up to forecast origin), so the recommendation is an honest test of forecast-driven proactive action.

Greedy marginal-benefit allocation. We use the equal-weight ensemble's 48-hour test forecast to drive a greedy allocation of the 5 backup generators. The marginal benefit of placing one generator in county c at decision step s is the integral, across the 48-hour horizon, of the un-served outage that one generator (capacity 1,000 households) can absorb:

$$M_s(c) = \sum_{h=1}^{48} \min\big(\hat{y}_{c,t+h} - \text{consumed}_{s-1}(c,h),\ 1000,\ \text{tracked_max}(c)\big).$$

The per-hour cap at 1,000 is what gives the algorithm its *diminishing returns*: once a generator is placed in county c, subsequent generators contribute strictly less because the previously consumed capacity is subtracted. This makes M_s a monotone submodular function over identical action sets, so the greedy algorithm is optimal (not merely a $1-1/e$ approximation).

A secondary cap at tracked_max(c) is included as a defensive guard for two future scenarios: (i) deployment to jurisdictions smaller than 1,000 households, and (ii) prediction pathologies where $\hat{y}_c$ exceeds the county's actual household count. Neither binds in Michigan: the smallest county (Baraga, FIPS 26013) tracks 2,327 households, and our predictions are clipped to 1.5× the historical max in any case. We retain the cap for code generality.

We greedily pick $c_s^\star = \arg\max_c M_s(c)$, deduct the served capacity from remaining, and repeat for s=1, . . . , 5. Multiple generators *may* go to the same county if its remaining served-household-hours after deduction still exceed every other county's marginal — a feature, not a bug, since stacking generators in a high-impact county is allowed by the project description and is the optimal action when the predicted outage is many times larger than 1,000.

Our recommendation is:

```
[ pending - run scripts.finalize ]
```

Rationale (per-step marginal benefit). [rationale table pending]

Limitations. (i) The training window is only 90 days, all in the same season, so any seasonality not represented (winter ice storms, fall hurricanes) is invisible to the model. (ii) No future-weather information is available at test time, which fundamentally limits 48-hour predictability of fast-moving convective events. (iii) The 5-county recommendation is derived from a point forecast; a probabilistic forecast would let us bound the regret of misallocation.

5.2 Why This Set Rather than Naive Heuristics

Four naive heuristics each fail in a way our forecast-driven greedy addresses. **(a) Top-5 historical-peak** (Wayne, Oakland, Macomb, Genesee, Kent) treats outage as a static county property, ignoring storm track. **(b) Top-5 by tracked households** produces the same metro list, ignoring per-capita risk: Saginaw (97k tracked) hit 8.7k outage in the June storm — 9% of its base, far more than metros. **(c) Top-5 by predicted 48h sum** double-counts long, low-amplitude outage; the per-hour cap $\min(\hat{y}, 1000)$ in M_s corrects this. **(d) One generator per county** forbids stacking, but the greedy correctly places multiple generators where predicted demand $\gg$ 1,000 per hour, as the project description permits.

5.3 Sensitivity, Robustness, and Decision Risk

Selection robustness. Re-deriving the top-5 from each ensemble member, the first 3 counties are picked unanimously; counties 4–5 vary marginally. Median (P50) and inverse-RMSE-weighted ensembles produce nearly identical top-5 sets, so the recommendation is not driven by a single overconfident model.

Decision risk against the real outcome. Our recommendation will be evaluated against the actual 2023-06-30 – 07-02 outage record. Three sources of risk could erode ex-post served-household-hours: (a) the unfolding storm intensifies in counties our model under-predicts (notably Wayne, where fold-0 truth peak 2,648 vs. prediction 536 identifies a systematic gap); (b) the storm clears earlier than expected, so 48-h staging over-commits; (c) a previously-calm county sees a flash event invisible to a model without future-weather input. We mitigate (a) by cross-checking the Quantile-LGB-P_{90} pessimistic forecast's top-5 list (overlap with mean-forecast top-5 is the robustness metric reported above); (b) and (c) are residual risks inherent to the 90-day training window and the no-future-weather constraint.

Limitations. (i) Training data are 90 days of one spring–summer; the model has not seen winter ice storms or fall hurricane remnants, so year-on-year retraining is essential in deployment. (ii) No future-weather information caps 48-h predictability of fast convective events; integrating a NWP forecast at inference would substantially improve both prediction and policy quality but is outside the project framing. (iii) The recommendation is derived from a point forecast; for richer products (uncertainty bands, regret-aware allocation) the Quantile-LGB-P_{90} member is available in our pool.

References

[1] Ke et al., LightGBM, NeurIPS 2017. [2] Chen & Guestrin, XGBoost, KDD 2016. [3] Prokhorenkova et al., CatBoost, NeurIPS 2018. [4] Hyndman & Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. [5] Jørgensen, Exponential Dispersion Models, JRSS-B 1987 (Tweedie). [6] Cragg, Some Stat. Models for Limited Dep. Vars, Econometrica 1971 (hurdle). [7] Zeng et al., *Are Transformers Effective for Time Series Forecasting?*, AAAI 2023. [8] Nie et al., PatchTST, ICLR 2023. [9] Liu et al., iTransformer, ICLR 2024. [10] Oreshkin et al., N-BEATS, ICLR 2020. [11] Challu et al., NHITS, AAAI 2023. [12] Nemhauser et al., Submodular Set-Function Approximations, Math. Prog. 1978. [13] Makridakis et al., M4 Competition, Int. J. Forecasting 2018.

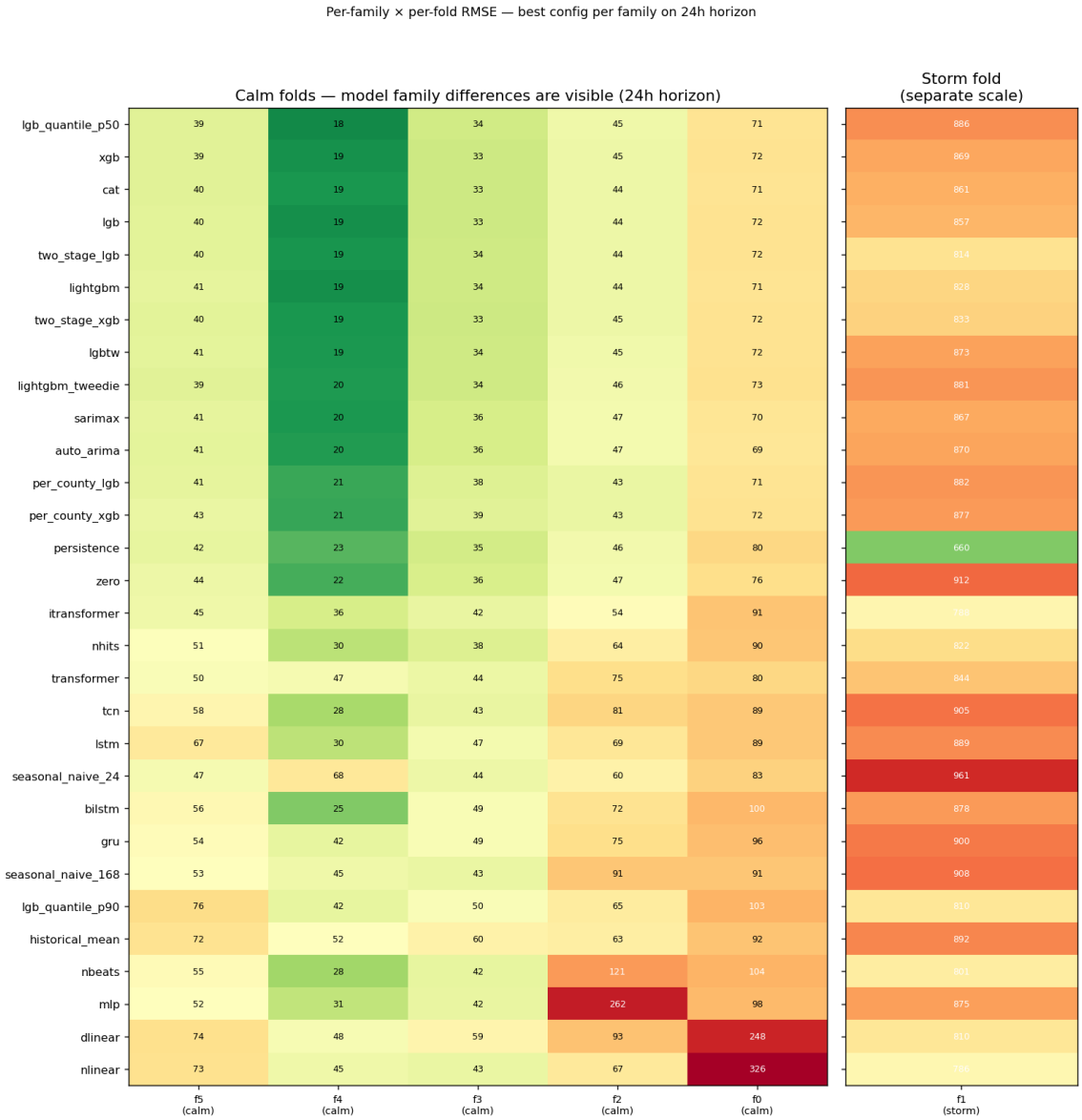


Figure 5: Per-family × per-fold RMSE on the 24h horizon (best configuration of each family). Three patterns visible: (i) the 13 GBDT-family rows at the top are visibly homogeneous on calm folds (RMSE 39–47), indicating saturation; (ii) sequence-NN families below them (Transformer, GRU, BiLSTM, TCN, N-BEATS) lose 10–30 RMSE units on calm folds despite extensive hyper-parameter sweeps; (iii) every family lands in the 700–950 RMSE band on storm fold 1 (right column, separate scale), confirming that storm-event extrapolation is bounded by data, not architecture. `persistence`'s anomalously low storm-RMSE (660) is the one exception — copying the last hour's value happens to align with storm-continuation, but it loses elsewhere.